

Twelfth Section. Theory of Roots of Unity.

§ 131. The Roots of Unity

By a root of unity one generally understands a real or imaginary number with the property that some power of it, with an integral exponent, is equal to 1. We have already met these roots of unity in § 33 and represented them by trigonometric and exponential functions. We have also used special cases, for example the third roots of unity in § 35, where they were expressed without trigonometric functions.

In all deeper investigations of algebraic equations, a more exact knowledge of the roots of unity and their properties is indispensable. In this section we shall therefore occupy ourselves more closely with the elementary part of the algebraic theory of roots of unity, without making use of their representation by trigonometric functions. Because of the geometric application to the construction of regular polygons, to which we already referred in § 33, the theory of roots of unity is also called the theory of division of the circle. In its essentials it is a creation of Gauss.¹

If the n th power of a number r is equal to 1, that is, if the equation

$$r^n = 1 \quad (1)$$

is satisfied for a positive integer n , then r is called an n th root of unity, or a root of unity of degree n . Thus all n th roots of unity, and only these, are roots of the equation of degree n

$$f(x) = x^n - 1 = 0. \quad (2)$$

Now

$$f'(x) = nx^{n-1}, \quad (3)$$

and consequently $f(x)$ has no common divisor with $f'(x)$. Hence $f(x)$ has no multiple roots, and there are exactly n distinct n th roots of unity.

If r is an n th root of unity, then so is every integral power of r ; for from $r^n = 1$ it follows that $r^{kn} = 1$ when k is any positive or negative integer, including $k = 0$. Thus r^k is also an n th root of unity. But since there are only n roots of unity of degree n , the powers r^k cannot all be different. The following facts hold.

If two integers k, k' differ by a multiple of n , that is, if

$$k' \equiv k \pmod{n}, \quad (4)$$

then also

$$r^{k'} = r^k. \quad (5)$$

Indeed, if $k' = k + hn$, then

$$r^{k'} = r^{k+hn} = r^k (r^n)^h,$$

and (5) follows from $r^n = 1$.

Thus the sequence

$$1, r, r^2, r^3, \dots, r^{n-1} \quad (6)$$

certainly contains all the distinct powers r^k ; but conversely the quantities in (6) need not all be distinct.

Suppose that h and $h' = h + \mu$ are two numbers in the sequence

$$0, 1, 2, \dots, n-1$$

¹Gauss, *Disquisitiones arithmeticae*, Sectio VII.

and that

$$r^h = r^{h'}.$$

It follows that $r^\mu = 1$. Hence no previously occurring term in the sequence (6) can recur before the first term 1 recurs for the second time; and if μ is the least positive number for which $r^\mu = 1$, then the numbers

$$1, r, r^2, \dots, r^{\mu-1} \quad (7)$$

are all distinct.

Then μ must be a divisor of n . For by division we can determine integers h, μ' such that

$$n = h\mu + \mu', \quad 0 \leq \mu' < \mu.$$

But then, from

$$r^n = (r^\mu)^h r^{\mu'},$$

one obtains $r^{\mu'} = 1$. Since μ is to be the least positive number for which $r^\mu = 1$, this gives $\mu' = 0$, or n is divisible by μ .

Thus r is at the same time a μ th root of unity, but not a root of unity of any still lower degree.

The number r is called a primitive n th root of unity if it is not also a root of unity of a lower degree.

It follows from this definition that the numbers in (6), when r is a primitive n th root of unity, are all distinct and contain all n th roots of unity; whereas, if r is an imprimitive n th root of unity, they make up only a part of the n th roots of unity.

Every root of unity whose degree is a divisor of n different from n is at the same time an imprimitive n th root of unity.

If r is simultaneously an n th and an m th root of unity, then it is also a μ th root of unity, where μ is the greatest common divisor of n and m . For by §118 one can determine integers x, y such that

$$mx + ny = \mu,$$

and consequently

$$r^\mu = r^{mx+ny} = (r^m)^x (r^n)^y = 1.$$

Corresponding to this is the following converse statement. If $r_1, r_2, \dots$ are roots of unity of degrees $n_1, n_2, \dots$, then they are all also roots of unity of degree m , where m denotes any common multiple of $n_1, n_2, \dots$.

§ 132. Primitive Roots of Unity

We have not yet proved in the preceding paragraph that primitive roots of unity exist for every degree n . We must first supply this proof, and at the same time we shall determine the exact number of primitive roots of unity.

Let the degree n be decomposed into two factors a, b which are relatively prime, so that

$$n = ab,$$

and let α be an a th root of unity and β a b th root of unity. Then the product

$$r = \alpha\beta \quad (1)$$

is an n th root of unity. If α', β' are two other a th and b th roots of unity, then $r' = \alpha'\beta'$ is also an n th root of unity; and one has to show that r' is different from r unless simultaneously $\alpha = \alpha'$ and $\beta = \beta'$.

Since a, b are relatively prime, we can, by § 118, determine integers x, y such that

$$ax + by = 1. \quad (2)$$

It follows from (1) that

$$\alpha = r^{by}, \quad \beta = r^{ax}.$$

Thus α and β are completely determined by r , and if $\alpha'\beta'$ is also equal to r , then necessarily $\alpha = \alpha'$ and $\beta = \beta'$.

Letting α run through all a th roots and β through all b th roots of unity in (1), one obtains exactly $ab = n$ distinct values of r . Hence:

I. all n th roots of unity are representable in the form $\alpha\beta$;

and further:

II. r is a primitive n th root of unity if and only if α is a primitive a th root and β is a primitive b th root of unity.

For first let μ be the smallest positive exponent for which $r^\mu = 1$. Then also $\alpha^\mu\beta^\mu = 1$, and, using (2) by raising both sides to the powers by and ax , one obtains

$$\alpha^\mu = 1, \quad \beta^\mu = 1.$$

If $\mu < n$, then μ cannot be divisible by both a and b ; consequently a and b cannot both be the least positive exponents of powers of α and β that become equal to 1. Thus if r is not a primitive n th root of unity, then α and β are not simultaneously primitive a th and b th roots. Conversely, if α and β are primitive roots of degrees a and b , then their product r is a primitive n th root. On the other hand it is clear that if α or β is at the same time a root of unity of degree lower than a or b , then r will also be a root of unity of degree lower than n .

If n decomposes into several factors $a, b, c, \dots$, any two of which are relatively prime, and if $\alpha, \beta, \gamma, \dots$ are roots of unity of degrees $a, b, c, \dots$, and one puts

$$r = \alpha\beta\gamma \cdots, \quad (3)$$

then repeated application of the preceding theorems gives that (3) contains all n th roots of unity, each once and only once; moreover, r is a primitive n th root if and only if $\alpha, \beta, \gamma, \dots$ are primitive roots of unity of degrees $a, b, c, \dots$

If we now denote by $\varphi(n)$ the number of primitive n th roots of unity, the preceding result gives

$$\varphi(n) = \varphi(a)\varphi(b)\varphi(c) \cdots, \quad (4)$$

when

$$n = abc \cdots$$

and $a, b, c, \dots$ are pairwise relatively prime.

Every number n can be decomposed in one and only one way into a product of powers of primes,

$$n = p^x p_1^{x_1} p_2^{x_2} \cdots,$$

where $p, p_1, p_2, \dots$ are distinct primes and $x, x_1, x_2, \dots$ are positive exponents. Formula (4) therefore gives

$$\varphi(n) = \varphi(p^x)\varphi(p_1^{x_1})\varphi(p_2^{x_2}) \cdots, \quad (5)$$

so that it remains only to decide whether, and how many, primitive roots of unity of degree p^x exist.

This question is very easy to decide. If q is a root of unity of degree p^x , then the lowest degree to which q belongs as a root of unity is a divisor of p^x , hence a power of p . If it is not equal to p^x ,

it is a divisor of p^{x-1} ; that is, every non-primitive root of unity of degree p^x is at the same time a root of unity of degree p^{x-1} . But there are p^x roots of unity of degree p^x and only p^{x-1} roots of unity of degree p^{x-1} . Thus there must be

$$p^x - p^{x-1} = p^x \left(1 - \frac{1}{p}\right)$$

primitive roots of unity of degree p^x . Hence, by (5), the number of all primitive n th roots of unity is

$$\varphi(n) = n \prod \left(1 - \frac{1}{p}\right), \quad (6)$$

where the product is to be taken over all distinct prime numbers dividing n . Only the case $n = 1$ no longer fits formula (6); in this case one sets $\varphi(1) = 1$.

The number $\varphi(n)$ is therefore never zero. Since consequently there is at least one primitive root of unity r for every degree n , all n th roots of unity can be represented by the powers of r :

$$1, r, r^2, r^3, \dots, r^{n-1}. \quad (7)$$

If r^k is any power of r , then $r^{km} = 1$ holds if and only if km is divisible by n . Thus, if $n = n'n''$ and n' is the greatest common divisor of k and n , then m must be divisible by n'' ; and n'' is the exponent of the lowest power of r^k that becomes equal to 1. Thus r^k is a primitive n'' th root of unity. Hence:

III. If r is a primitive n th root of unity, then r^k is a primitive n th root of unity if and only if k is relatively prime to n .

Taking k from the sequence $1, 2, 3, \dots, n$, one obtains the theorem of number theory that $\varphi(n)$ is equal to the number of integers not greater than n and relatively prime to n . This is the original definition of the symbol $\varphi(n)$, which is in general use in number theory.

If k is relatively prime to n , then an integer x can be determined so that $kx \equiv 1 \pmod{n}$. If r, r_1 are two n th roots of unity, then $r^k = r_1^k$ can hold only if $r = r_1$, as is seen by raising to the power x . By III this gives:

IV. If k is relatively prime to n and r runs through the primitive n th roots of unity, then r^k runs through the same sequence of numbers, although in another order.

Finally we record:

V. If n is a prime number, then every n th root of unity except 1 is a primitive n th root of unity.

§ 133. Equations for the Primitive Roots of Unity of Degree n

All n th roots of unity are, as we have seen, roots of an equation $f_n(x) = 0$, where

$$f_n(x) = x^n - 1. \quad (1)$$

If we remove from the function $f_n(x)$ all factors that it has in common with other functions of the same form $f_\mu(x)$, which is done by rational operations (§ 6), then we obtain an equation

$$X_n = 0$$

satisfied by the primitive n th roots of unity and only by them; and X_n has the form

$$X_n = x^\nu + a_1 x^{\nu-1} + \dots + a_\nu. \quad (2)$$

The degree ν is equal to $\varphi(n)$, and the coefficients $a_1, a_2, \dots, a_\nu$ are rational numbers.

In looking for the common factors of f_n and f_μ , we may restrict μ to the divisors of n . We shall soon see more closely how the function X_n can be formed simply; first we prove a general theorem about these functions.

The n th roots of unity comprise all primitive μ th roots of unity, where μ is any divisor of n , including n itself and 1, since the primitive first root of unity is simply the unit 1 itself. Thus, if μ is allowed to run through all divisors of n , and if we observe that two different X_μ never have a common divisor, and that neither $f_n(x)$ nor any X_μ has multiple factors, we get

$$f_n(x) = \prod_{\mu|n} X_\mu. \quad (3)$$

This also gives a proof of the number-theoretic theorem

$$n = \sum_{\mu|n} \varphi(\mu), \quad (4)$$

by equating the degrees of the functions on the right and on the left in (3).

On the other hand, by Gauss's theorem (§2), we conclude that the coefficients of all the X_μ are integers. For if fractional coefficients occurred among them, the product could not have only integral coefficients, as it must by (3) and (1).

All functions $f_n(x)$ have the divisor $x - 1$, and carrying out the division gives

$$\frac{x^n - 1}{x - 1} = x^{n-1} + x^{n-2} + \dots + x + 1. \quad (5)$$

It follows that if r is any primitive or imprimitive n th root of unity, with the sole exception of $r = 1$, then

$$1 + r + r^2 + \dots + r^{n-1} = 0, \quad (6)$$

whereas for $r = 1$ the sum on the left side of (6) plainly has the value n .

If n is a prime number, then there are no imprimitive n th roots of unity except 1; and therefore, when n is prime, which we indicate by writing p for it,

$$X_p = x^{p-1} + x^{p-2} + \dots + x + 1. \quad (7)$$

Equally simple is the formation of X_n when n is a power of a prime, say

$$n = p^\pi,$$

or, putting $p^{\pi-1} = p'$ for abbreviation,

$$n = p'p.$$

The n th roots of unity then consist of the primitive n th roots and the p' th roots of unity; hence

$$X_n = \frac{x^{p'p} - 1}{x^{p'} - 1} = x^{p'(p-1)} + x^{p'(p-2)} + \dots + x^{p'} + 1. \quad (8)$$

If $p' > 1$, hence $\pi > 1$, then this equation lacks the term with the $(\nu - 1)$ st power of the unknown, whose coefficient is equal to the negative of the sum of the roots. Thus:

The sum of all primitive n th roots of unity is always zero when n is a higher power of a prime number.

For the general formation of X_n we shall use a recurrence procedure. We suppose X_n already formed, denote by p a prime not dividing n , by p' as above the $(\pi - 1)$ st power of p , and now form $X_{npp'}$, where of course p' may also be equal to 1.

Let r denote the primitive n th roots of unity, let α be any root of unity of degree pp' , and α' any root of unity of degree p' . Then all primitive roots of unity of degree npp' are obtained by taking all $r\alpha$ and removing the $r\alpha'$ (§132, II). But the $r\alpha$ are the roots of the equation

$$X_n(x^{pp'}) = 0,$$

since the quantities $(r\alpha)^{pp'} = r^{pp'}$, apart from their order, agree with the r themselves (§132, IV). Similarly, the $r\alpha'$ are the roots of the equation

$$X_n(x^{p'}) = 0.$$

Thus

$$X_{npp'}(x) = \frac{X_n(x^{pp'})}{X_n(x^{p'})}. \quad (9)$$

The value $n = 1$ is no exception if by X_1 we understand the function $x - 1$. Thus X_n can be formed in all cases in a relatively simple manner.

We consider a few special cases. Suppose first that n contains only two distinct prime numbers p, q , say

$$n = p^\pi q^\chi.$$

Then formulas (8) and (9) give

$$X_n = \frac{(x^n - 1)(x^{n/pq} - 1)}{(x^{n/p} - 1)(x^{n/q} - 1)}.$$

This formula is easily generalized by complete induction as follows. Let n_1 denote all numbers obtained from n by dividing it by an even number of distinct prime divisors of n (including n itself), and let n_2 denote all numbers obtained from n by dividing it by an odd number of such prime divisors. Then

$$X_n = \frac{\prod (x^{n_1} - 1)}{\prod (x^{n_2} - 1)}. \quad (10)$$

The proof follows from (9) for npp' , assuming the formula already proved for n .

If r denotes any primitive n th root of unity, then for odd n the numbers $-r$ are the primitive $2n$ th roots of unity (§132, I, II). Hence, for odd n ,

$$X_{2n}(x) = X_n(-x). \quad (11)$$

If n is a power of 2, then by (8)

$$X_n = \frac{x^n - 1}{x^{n/2} - 1} = x^{n/2} + 1. \quad (12)$$

Putting $x = 1$ in formula (8), X_n takes the value p when n is a power of p . On the other hand, formula (9), for $n > 1$, gives the value 1 for $X_{npp'}$; for $n = 1$ numerator and denominator on the right would both be zero. Thus:

V. The function X_n takes the value p for $x = 1$ when n is a power of the prime p , and the value 1 when n has more than one prime divisor.

§ 134. Irreducibility

The functions X_n defined in the preceding paragraph possess the property of irreducibility, which is very important for their deeper study and for all applications; that is:

It is not possible to decompose a function X_n of x into factors of lower degree which are themselves integral rational functions of x with rational numerical coefficients.

This important theorem was stated and first proved by Gauss for the case in which n is prime. Many other proofs were later given, some of them more general and simpler. We follow here a proof of Kronecker.

In the function X_n , as we saw above, the highest power of x has coefficient 1, while the remaining coefficients are integers. Suppose now that X_n can be decomposed into two factors with rational coefficients,

$$X_n = \varphi(x)\psi(x). \quad (1)$$

Then the product of the coefficients of the highest powers of x in φ and ψ is equal to 1, and we may therefore assume that these leading coefficients in φ and ψ themselves have the value 1. It follows from Gauss's theorem (§2) that the remaining coefficients of $\varphi(x)$ and $\psi(x)$ must be integers; hence $\varphi(x)$ and $\psi(x)$ take integer values when an integer is substituted for x .

Putting $x = 1$, we obtain

$$\varphi(1)\psi(1) = X_n(1),$$

which, by theorem V, is equal either to 1 or to a prime number p . Thus certainly one of the two factors $\varphi(1), \psi(1)$ has the value ± 1 , while the other is equal to ± 1 or $\pm p$. We may therefore assume that

$$\varphi(1) = \pm 1. \quad (2)$$

The roots r of the equation $X_n = 0$ are distributed between the two equations $\varphi = 0$ and $\psi = 0$. Since neither of the two factors φ, ψ is independent of x , there must be among the primitive n th roots of unity one, which we denote by ρ , for which

$$\varphi(\rho) = 0.$$

If r now denotes any primitive n th root of unity, then among the powers

$$r, r^2, r^3, \dots, r^{n-1}$$

all n th roots of unity except 1 occur, hence also ρ . It follows that the product

$$\Theta(x) = \varphi(x)\varphi(x^2)\varphi(x^3) \cdots \varphi(x^{n-1}) \quad (3)$$

vanishes for $x = r$. Since r can be any root of X_n , and these roots are all distinct, $\Theta(x)$ must be divisible by X_n , say

$$\Theta(x) = X_n\Psi(x). \quad (4)$$

The function $\Theta(x)$, by its construction, has coefficient 1 at its highest power of x and otherwise integral coefficients; the same is true of X_n . Therefore $\Psi(x)$ also must have coefficient 1 at its highest power of x and otherwise integral coefficients. But from (2) and (3), $\Theta(x)$ takes the value ± 1 for $x = 1$, so that

$$\pm 1 = X_n(1)\Psi(1), \quad (5)$$

where $\Psi(1)$ must be an integer.

If n is a prime power p^π , then $X_n(1) = p$, contradicting (5). Hence the irreducibility of X_n is proved for the case in which n is a prime, or a power of a prime.

The general proof still requires the following. If a and b are two relatively prime integers, and if X_a and X_b are assumed irreducible, then X_{ab} is also irreducible.

To prove this, put $n = ab$ and suppose that $\varphi(x)$ is a rational factor of X_n . The roots of $\varphi(x) = 0$ are then among the primitive n th roots of unity, and hence are of the form

$$\rho = \alpha\beta, \quad (6)$$

where α and β denote certain primitive roots of unity of degrees a and b (§132).

From $\varphi(x)$ one can now derive, by a special case of the Tschirnhausen transformation, a function $\Theta(x)$ with rational coefficients whose roots are the a th powers of the roots of $\varphi(x)$. The roots of $\Theta(x)$ are therefore

$$\rho^a = \beta^a,$$

that is, since a is relatively prime to b , primitive b th roots of unity. Thus, if any ρ is present at all, $\Theta(x)$ must have a non-constant factor in common with X_b . But we have assumed it already proved that X_b has no rational factor except itself. Consequently $\Theta(x)$ must be divisible by X_b , and the β^a , and hence also the β themselves, that occur in (6), comprise all primitive b th roots of unity. In the same way one proves that α in (6) runs through all primitive a th roots of unity, and that therefore ρ runs through all primitive n th roots of unity.

Thus the roots of $\varphi(x)$ completely coincide with all roots of X_n . Consequently, if to determine the numerical factor we also prescribe that the highest power of x in $\varphi(x)$ have coefficient 1, then $\varphi(x)$ must be identical with X_n .

Under the stated assumptions, therefore, X_n is not decomposable into factors of lower degree with rational coefficients.

With the same auxiliary tools we can prove the following theorem of Kronecker, which asserts the irreducibility of the functions X_n in a wider sense.²

The function X_n cannot be decomposed into factors that are integral and rational with respect to x , and whose coefficients contain only rational numbers and roots of unity whose degree is relatively prime to n .

Assume that a is relatively prime to n , that r is a primitive n th root and α a primitive a th root of unity, and that $\varphi(x, \alpha)$ is a divisor of X_n , so that for at least one r

$$\varphi(r, \alpha) = 0. \quad (7)$$

Let ρ be a primitive n th root of unity, hence a root of $X_n = 0$. Then both r and α are powers of ρ (§ 131), say

$$r = \rho^\xi, \quad \alpha = \rho^\eta,$$

where ξ is relatively prime to n but divisible by a , and η is relatively prime to a but divisible by n . Thus

$$\varphi(\rho^\xi, \rho^\eta) = 0. \quad (8)$$

Now, as just proved, X_n is irreducible, and therefore the function $\varphi(x^\xi, x^\eta)$, which has a common divisor with X_n , must be divisible by this function. Hence for every integer h relatively prime to na ,

$$\varphi(\rho^{h\xi}, \rho^{h\eta}) = 0. \quad (9)$$

If s is any integer relatively prime to n , then h can be chosen so that

$$h \equiv s \pmod{n}, \quad h \equiv 1 \pmod{a}.$$

It follows from (9) that

$$\varphi(r^s, \alpha) = 0,$$

that is, equation (7) is satisfied for all roots of X_n . Hence $\varphi(x, \alpha)$ is divisible by X_n and, since conversely X_n was assumed divisible by $\varphi(x, \alpha)$, it must agree with X_n up to a numerical factor.

²Kronecker, Mém. sur les facteurs irréductibles de l'expression $x^n - 1$, Liouville's Journal, Vol. XIV.

§ 135. The Discriminant of the Cyclotomic Equation

The roots of unity of degree n can be represented in transcendental form by

$$r = e^{2\pi i k/n} = \cos \frac{2\pi k}{n} + i \sin \frac{2\pi k}{n}. \quad (1)$$

This root of unity is primitive or imprimitive according as k is or is not relatively prime to n . The simplest primitive one is obtained for $k = 1$, namely

$$r_0 = e^{2\pi i/n} = \cos \frac{2\pi}{n} + i \sin \frac{2\pi}{n} = a + bi. \quad (2)$$

The angle $2\pi : n$ is the n th part of the whole circumference of the circle. The different points of division obtained by laying off this angle on the circumference from an arbitrary starting point determine the regular n -gon inscribed in the circle.

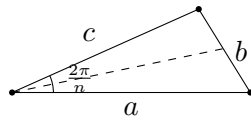


Fig. 28. Triangle for the side of the regular n -gon.

The points of division can be constructed if one knows r and hence a and b , or even if one knows only one of these two quantities. In particular, for the side s of the regular n -gon, when the radius c of the circumscribed circle is taken equal to 1, one obtains

$$s = 2 \sin \frac{\pi}{n} = \sqrt{2 \left(1 - \cos \frac{2\pi}{n} \right)}.$$

The other quantities r stand in the same relation to the other division points. Because of this geometric meaning, we call the equation $X_n = 0$, whose roots are the quantities (1), the cyclotomic equation.

We now determine the discriminant of the cyclotomic equation $X_n = 0$, but restrict ourselves to the case in which n is an odd prime number. Then

$$X_n = x^{n-1} + x^{n-2} + \cdots + x + 1, \quad (3)$$

and since the only imprimitive n th root of unity is 1,

$$f_n(x) = x^n - 1 = (x - 1)X_n, \quad (4)$$

so that

$$f'_n(x) = nx^{n-1} = X_n + (x - 1)X'_n. \quad (5)$$

If D is the discriminant of X_n , then by § 46

$$D = (-1)^{(n-1)/2} \prod X'_n(r), \quad (6)$$

where the product extends over all roots r of the equation $X_n = 0$.

But by (5)

$$nr^{n-1} = (r - 1)X'_n(r), \quad (7)$$

and since the product of all r is equal to 1 (because it equals the constant term of X_n), and since

$$\prod (r - 1) = X_n(1) = n,$$

we obtain from (6) and (7)

$$D = (-1)^{(n-1)/2} n^{n-2}. \quad (8)$$

If r is one of the roots of X_n , then all its roots are contained in the form

$$r, r^2, r^3, \dots, r^{n-1}.$$

Introducing the product of differences

$$P = (r - r^2)(r - r^3) \dots (r - r^{n-1}) \cdot (r^2 - r^3) \dots (r^2 - r^{n-1}) \dots (r^{n-2} - r^{n-1}), \quad (9)$$

we have by §46

$$D = P^2.$$

Thus also, by (8),

$$P = n^{(n-2)/2} \sqrt{(-1)^{(n-1)/2}}, \quad (10)$$

which determines P up to sign. It is real when $n \equiv 1 \pmod{4}$ and imaginary when $n \equiv -1 \pmod{4}$, hence real for $n = 5, 13, 17, 29, 37, \dots$ and imaginary for $n = 3, 7, 11, 19, 23, 31, \dots$. The sign to be attached to the root in (10) depends on which r is chosen in the expression (9).

If we choose a definite r , for example $r = r_0$, then we can still determine the sign in (10). For this purpose we divide the binomial factors of P , whose number is $n(n-1)/2$, into two classes. The differences of one class form the product

$$Q = (r - r^{n-1})(r^2 - r^{n-2}) \dots (r^{(n-1)/2} - r^{(n+1)/2}), \quad (11)$$

which therefore contains all factors for which $\mu + \nu = n$. The remaining factors can be combined in pairs of the form

$$R = (r^\mu - r^\nu)(r^{n-\mu} - r^{n-\nu}). \quad (12)$$

The number of factors in Q is $(n-1)/2$, and hence the number of pairs R is

$$\frac{1}{2} \left(\frac{n(n-1)}{2} - \frac{n-1}{2} \right) = \frac{(n-1)^2}{4}.$$

Now, if

$$r = e^{2\pi i k/n},$$

then

$$R = -2 + r^{\mu-\nu} + r^{\nu-\mu} = -2 \left(1 - \cos \frac{2(\mu-\nu)k\pi}{n} \right),$$

so that R is always negative. Consequently the product of all R has the sign

$$(-1)^{(n-1)^2/4}.$$

Furthermore

$$Q = (2i)^{(n-1)/2} \sin \frac{2\pi k}{n} \sin \frac{4\pi k}{n} \dots \sin \frac{(n-1)\pi k}{n}, \quad (13)$$

and its sign depends on k . If however $k = 1$, so that $r = r_0$, then all the angles

$$\frac{2\pi}{n}, \quad \frac{4\pi}{n}, \quad \dots, \quad \frac{(n-1)\pi}{n}$$

lie between 0 and π , and all the sines are positive. Hence in this case

$$P = (-i)^{(n-1)^2/4} n^{(n-2)/2} \sqrt{n}, \quad (14)$$

with $\sqrt{n}$ to be taken positive.

The sign of the product of the R does not depend on the choice of r , but only on the nature of n . It is therefore of interest to determine separately the product Q , whose sign does depend on the choice of r . We obtain it as follows.

Multiply the expression (11) by

$$r^{1+2+\dots+(n-1)/2}$$

and then by

$$r^{-1}r^{-2}\dots r^{-(n-1)/2}.$$

This gives

$$\begin{aligned} r^{(n^2-1)/8}Q &= (r^2-1)(r^4-1)\dots(r^{n-1}-1), \\ r^{-(n^2-1)/8}Q &= (1-r^{n-2})(1-r^{n-4})\dots(1-r). \end{aligned}$$

Since the exponents $2, 4, \dots, n-1, n-2, n-4, \dots, 1$ together comprise all numbers $1, 2, \dots, n-1$, multiplication of these two expressions gives

$$Q^2 = (-1)^{(n-1)/2} \prod (r-1) = (-1)^{(n-1)/2} n,$$

so that

$$Q = \pm i^{(n-1)/2} \sqrt{n}. \quad (15)$$

Comparison with (13) gives

$$2^{(n-1)/2} \sin \frac{2\pi k}{n} \sin \frac{4\pi k}{n} \dots \sin \frac{(n-1)\pi k}{n} = \pm \sqrt{n}. \quad (16)$$

The sign in this formula still depends on k , as in (15), but it is positive when $k=1$. We shall now derive a further important theorem about this sign, which will be investigated more closely later.

By definition (11), Q is an integral rational function of r which, when arranged according to powers of r , has integral rational numerical coefficients. Thus, if

$$r = r_0^k,$$

then

$$Q = F(r_0^k),$$

where F denotes a rational function whose coefficients do not depend on k . Now putting successively

$$k = 1, 2, \dots, n-1,$$

the values r_0^k represent all roots of the cyclotomic equation $X_n = 0$, and the sum

$$\sum Q = F(r_0) + F(r_0^2) + \dots + F(r_0^{n-1})$$

is a symmetric function of the roots of this equation. It can therefore be expressed rationally by the coefficients of the equation, that is, by rational numbers, and is itself a rational number. On the other hand, by formula (15),

$$\sum Q = h i^{(n-1)/2} \sqrt{n},$$

where h is an integer, namely the number of cases in which the positive sign is to be taken in (15), diminished by the number of cases in which the negative sign holds. These two facts are compatible only if $h=0$. This proves the following theorem:

1. As k runs through the numbers $1, 2, 3, \dots, n-1$, the positive and the negative signs occur equally often in formula (16).

It scarcely needs to be mentioned that for k one may also take any other complete system of residues modulo n , excluding the number divisible by n .

§ 136. Primitive Congruence Roots

Parallel with the theory of roots of unity there runs a theory of the so-called binomial congruences. Without it no further steps can be taken in the theory of cyclotomic equations; its elements are suitably inserted here, where the analogy with roots of unity is clear.

Let now n be a prime number and

$$f(x) = a_0x^m + a_1x^{m-1} + \cdots + a_{m-1}x + a_m \quad (1)$$

an integral function of x , whose coefficients a are integers, and with a_0 not divisible by n . If we substitute for x an integer α such that $f(\alpha)$ is divisible by n , then, by analogy with equations, we say that α is a root of the congruence of degree m

$$f(x) \equiv 0 \pmod{n}. \quad (2)$$

If α is increased by a multiple of n , it remains a root of (2). Roots congruent modulo n are not counted as different. With this convention we can state:

1. A congruence of degree m modulo a prime number cannot have more than m different roots.

The theorem is true for $m = 1$, since the congruence $a_0x + a_1 \equiv 0 \pmod{n}$ has only one root, namely, if a'_0 is chosen so that $a_0a'_0 \equiv 1 \pmod{n}$, $x \equiv -a'_0a_1 \pmod{n}$. We now assume the theorem proved for degree $m - 1$ and derive its truth for degree m . By § 4, with α initially arbitrary, we can write

$$f(x) - f(\alpha) = (x - \alpha)f_1(x), \quad (3)$$

where $f_1(x)$ is an integral function of degree $m - 1$ which, when α is an integer, has integral coefficients. Hence, if $f(\alpha) \equiv 0$, (3) gives

$$f(x) \equiv (x - \alpha)f_1(x) \pmod{n}. \quad (4)$$

Every root β of (2) must therefore satisfy $(\beta - \alpha)f_1(\beta) \equiv 0 \pmod{n}$. Thus either $\beta \equiv \alpha$, or $f_1(\beta)$ is divisible by n .

By the induction hypothesis there are at most $m - 1$ values β for which $f_1(\beta)$ is divisible by n . If a further m th root of (4) exists, it must therefore be equal to α .

Thus if a congruence of the form $f(x) \equiv 0 \pmod{n}$ has more roots than its degree, the congruence must be identical; that is, all coefficients of $f(x)$ must be divisible by n .

This theorem is plainly analogous to the algebraic theorem that an equation cannot have more roots than its degree. But the other theorem, that every equation actually has as many roots as its degree, cannot be transferred. A congruence of degree m can have fewer roots, or even none. It is therefore all the more notable that there is a special congruence for which the number of roots always equals the degree. This rests on the theorem called Fermat's theorem, which we formulate as follows.

2. The congruences

$$x^n - x \equiv 0, \quad x^{n-1} - 1 \equiv 0 \pmod{n} \quad (5)$$

have, when n is prime, as many roots as their degrees, namely n and $n - 1$.

The two assertions are not essentially different, since the first congruence in (5) has all roots of the second and in addition the root 0, which does not satisfy the second. Likewise, the second has all roots of the first except 0.

Since modulo n there are altogether only n different numbers, it remains to prove that for every integer a the congruence

$$a^n \equiv a \pmod{n} \quad (6)$$

holds. This congruence is true for $a = 0$ and $a = 1$. We prove it again by complete induction, deriving from the assumed truth of (6) the truth of

$$(a + 1)^n \equiv a + 1 \pmod{n}. \quad (7)$$

This follows from the binomial theorem, since all binomial coefficients except the first and the last, which are equal to 1, namely

$$n, \quad \frac{n(n-1)}{1 \cdot 2}, \quad \frac{n(n-1)(n-2)}{1 \cdot 2 \cdot 3}, \dots,$$

are integers divisible by n , because the numerator but not the denominator is divisible by n . Hence

$$(a + 1)^n \equiv a^n + 1 \pmod{n},$$

and formula (7) follows from formula (6).

The difference

$$x(x-1)(x-2) \cdots (x-n+1) - x^n + x$$

is a function of degree at most $n-1$. But for n values of x , namely $x = 0, 1, 2, \dots, n-1$, it is congruent to zero. Therefore identically

$$x(x-1)(x-2) \cdots (x-n+1) \equiv x^n - x \pmod{n}.$$

From this follows Wilson's theorem,

$$1 \cdot 2 \cdot 3 \cdots (n-1) \equiv -1 \pmod{n},$$

by comparing the coefficients of the first power of x on both sides.

We now restrict ourselves to the second of the congruences (5),

$$x^{n-1} \equiv 1 \pmod{n}, \quad (8)$$

and first prove:

3. If a is a divisor of $n-1$, then the congruence

$$x^a \equiv 1 \pmod{n} \quad (9)$$

has exactly a different roots.

For if $n-1 = ab$, then

$$x^{n-1} - 1 = (x^a - 1)(x^{a(b-1)} + x^{a(b-2)} + \cdots + x^a + 1),$$

and since every root of the left side must be a root of one or the other factor on the right, it follows that if $x^a - 1 \equiv 0$ had fewer than a roots, the second factor would have to have more roots than its degree, contrary to theorem 1.

If α is not divisible by n , and a is the least positive number for which $\alpha^a \equiv 1 \pmod{n}$, and if h is any positive integer for which $\alpha^h \equiv 1 \pmod{n}$, then a is necessarily a divisor of h , hence in particular a divisor of $n-1$. For if $h = qa + a'$ with q an integer and $a' < a$, then also $\alpha^{a'} \equiv 1$; hence $a' = 0$, since a was the least positive number for which this congruence is satisfied.

A root α of the congruence (9) with the property that no lower power than the a th is congruent to unity is called a primitive root of this congruence, or a primitive a th congruence root of the

prime number n . A primitive root g of the congruence (8) is also called simply a primitive root of the prime n .

We shall prove that primitive roots exist for every prime n and determine their number, following the same route as for roots of unity.

Let a and b be two divisors of $n - 1$, relatively prime to each other, and let α be a primitive a th and β a primitive b th congruence root modulo n . Put $ab = c$. Then $\gamma = \alpha\beta$ is a primitive c th congruence root; and conversely every primitive c th congruence root can be represented in the form $\alpha\beta$.

Indeed, first $(\alpha\beta)^c = \alpha^c\beta^c \equiv 1 \pmod{n}$. Secondly, if $(\alpha\beta)^h = \alpha^h\beta^h \equiv 1 \pmod{n}$, then also $\alpha^{hb}\beta^{hb} \equiv 1$, hence $\alpha^{hb} \equiv 1$; so hb is divisible by a , and therefore h is divisible by a . Similarly h is divisible by b , hence by $ab = c$. Thus $\alpha\beta$ is a primitive c th congruence root.

Conversely, if γ is a primitive c th congruence root, determine positive integers x, y so that

$$ay + bx \equiv 1 \pmod{c} \quad (10)$$

by first determining x from $bx \equiv 1 \pmod{a}$ and then y from $ay \equiv 1 - bx \pmod{b}$. Then y is relatively prime to b and x relatively prime to a . Hence

$$\gamma = \gamma^{ay+bx} = \alpha\beta \pmod{n},$$

and $\gamma^{bx} = \alpha$ is a primitive a th congruence root, $\gamma^{ay} = \beta$ a primitive b th congruence root; for $\alpha^h = \gamma^{bhx}$ can be equal to 1 only if h is divisible by a .

It remains to show that the products $\alpha\beta$ are all different, that is, that the congruence

$$\alpha\beta \equiv \alpha'\beta' \pmod{n} \quad (11)$$

where α, α' and β, β' are primitive a th and b th congruence roots, can be satisfied only when

$$\alpha \equiv \alpha', \quad \beta \equiv \beta' \pmod{n}.$$

This follows from (10) by raising (11) to the power $bx \equiv 1 \pmod{a}$, which gives $\alpha \equiv \alpha'$, and then also $\beta \equiv \beta'$.

If therefore the number of primitive a th congruence roots is denoted by $\varphi(a)$, what has just been proved gives

$$\varphi(c) = \varphi(a)\varphi(b). \quad (12)$$

It remains only, if p is a prime and p^x is a power of p dividing $n - 1$, to determine $\varphi(p^x)$. If α is an imprimitive congruence root of degree p^x , and $p^{x-1} = p_1$, then, if α^{p_1} is the lowest power of α congruent to 1 modulo n , p_1 is a divisor of pp_1 , hence a power of p ; but since p_1 is to be smaller than pp_1 , it is also a divisor of p_1 , and consequently $\alpha^{p_1} \equiv 1 \pmod{n}$. Thus a non-primitive congruence root of degree pp_1 is also a congruence root of degree p_1 . Conversely, every congruence root of degree p_1 is at the same time an imprimitive congruence root of degree pp_1 . Since the congruences $x^{pp_1} \equiv 1$ and $x^{p_1} \equiv 1$ have as many roots as their degrees, there remain $pp_1 - p_1$ primitive congruence roots of the first, and therefore

$$\varphi(pp_1) = p_1(p - 1). \quad (13)$$

The function $\varphi(a)$ thus has the same meaning as in §132, and the number of primitive roots of the prime n is thereby found to be $\varphi(n - 1)$. We state the result:

4. A prime number n has $\varphi(n - 1)$ primitive roots.

Thus the existence of primitive roots for every prime is proved, and at the same time their number is determined. For finding primitive roots, however, we have no general method except trial, which can be shortened somewhat by a few artifices.

If g is a primitive root of the prime n , then the residues of the powers

$$1, g, g^2, g^3, \dots, g^{n-2} \quad (14)$$

are all distinct; for if $g^h \equiv g^{h+\mu}$, then $g^\mu \equiv 1$, which is impossible so long as μ is smaller than $n-1$. Hence among the residues of (14), which do not include the residue 0, every one of the numbers

$$1, 2, 3, \dots, n-1 \quad (15)$$

occurs once and only once. In other words:

5. If a is not divisible by n , then there is one and only one number α from the sequence $0, 1, 2, \dots, n-2$ by which the congruence

$$g^\alpha \equiv a \pmod{n} \quad (16)$$

is satisfied. The same congruence is also satisfied when an exponent congruent to α modulo $n-1$ is substituted, and in every complete residue system one finds only one such number α .

The exponent α is called the index of a with respect to the base g , and one writes

$$\alpha \equiv \text{ind } a \pmod{n-1}. \quad (17)$$

Thus for every number not divisible by n ,

$$g^{\text{ind } a} \equiv a \pmod{n}. \quad (18)$$

From the two formulas

$$g^{\text{ind } a + \text{ind } b} \equiv ab, \quad g^{m \text{ ind } a} \equiv a^m \pmod{n} \quad (19)$$

follows:

6. The index of a product is equal to the sum of the indices of the factors; the index of the m th power of a is equal to m times the index of a .

These theorems are exactly analogous to the corresponding theorems for calculation with logarithms, and the analogy can be pursued further. For example, passage from one base g to another base g' is effected by the formula

$$\text{ind}_g a \equiv \text{ind}_g g' \text{ ind}_{g'} a \pmod{n-1}. \quad (20)$$

We also mention the easily proved theorem that, among numbers a not divisible by n , precisely the primitive roots are those whose indices are relatively prime to $n-1$.

For practical computations one conveniently uses so-called index tables, corresponding to tables of logarithms. In the congruence (17), α is called the index and a the numerus. It is best to set up two tables: one giving the numerus for each index, the other giving the index for each numerus; the second is obtained from the first by rearrangement. Here one must observe that the modulus for the indices is $n-1$, while the modulus for the numeri is n , and congruent numbers count as equivalent. Thus, for $n=13$ and base 2, one obtains

I	0	1	2	3	4	5	6	7	8	9	10	11
N	1	2	4	8	3	6	12	11	9	5	10	7
N	1	2	3	4	5	6	7	8	9	10	11	12
I	0	1	4	2	9	5	11	3	8	10	7	6

In Jacobi's *Canon Arithmeticus* (1839), index tables are collected for all prime numbers in the first thousand. On a smaller scale, tables after computations of Ostrogradsky are included in Tschebyscheff's *Theorie der Congruenzen* (German by Schapira, Berlin 1889).

It remains to note that the index of 1 is always equal to 0, and that the index of -1 (or of $n-1$) is always equal to $(n-1)/2$. For since $(-1)^2$ has index 0 or $n-1$, -1 cannot have index 0, and since twice its index is equal to 0 or $n-1$, it must have index $(n-1)/2$. In formulas,

$$\text{ind } 1 = 0, \quad \text{ind}(-1) \equiv \frac{n-1}{2} \pmod{n-1}. \quad (21)$$

§ 137. Multiplication and Division of the Trigonometric Functions

We have already pointed out the connection between the theory of roots of unity and the geometric problem of dividing the circumference of a circle. The generalization of this problem leads to the division of an arbitrary angle. This problem too leads to algebraic equations; we have already used the equation for trisection in the general solution of cubic equations (§ 112). We now begin by setting up the general formulas.

By de Moivre's theorem (§ 33), if n is any positive integer and φ any angle,

$$\cos n\varphi + i \sin n\varphi = (\cos \varphi + i \sin \varphi)^n. \quad (1)$$

Applying the binomial theorem to the right side and separating real and imaginary parts gives

$$\begin{aligned} \cos n\varphi &= \cos^n \varphi - B_2^{(n)} \cos^{n-2} \varphi \sin^2 \varphi + B_4^{(n)} \cos^{n-4} \varphi \sin^4 \varphi - \dots, \\ \frac{\sin n\varphi}{\sin \varphi} &= n \cos^{n-1} \varphi - B_3^{(n)} \cos^{n-3} \varphi \sin^2 \varphi + B_5^{(n)} \cos^{n-5} \varphi \sin^4 \varphi - \dots. \end{aligned} \quad (2)$$

Here $B_\nu^{(n)}$ denotes the binomial coefficients as before, and the sums on the right are to be continued as long as no negative exponents of $\cos \varphi$ occur. Since $\sin^2 \varphi = 1 - \cos^2 \varphi$, both expressions can be represented rationally by $\cos \varphi$. Put

$$2 \cos \varphi = x, \quad (3)$$

and introduce the notation

$$2 \cos n\varphi = A_n(x), \quad \frac{\sin n\varphi}{\sin \varphi} = B_n(x). \quad (4)$$

Thus $A_n(x)$ and $B_n(x)$ are integral rational functions of x , with $A_n(x)$ of degree n and $B_n(x)$ of degree $n - 1$. We shall determine their general law. First we write them for $n = 1, 2, 3, \dots$, and then derive a recurrence formula for computing the higher functions. One finds

$$\begin{aligned} A_1(x) &= x, & B_1(x) &= 1, \\ A_2(x) &= x^2 - 2, & B_2(x) &= x, \\ A_3(x) &= x^3 - 3x, & B_3(x) &= x^2 - 1. \end{aligned} \quad (5)$$

But by the known formulas

$$\begin{aligned} \cos(n+1)\varphi + \cos(n-1)\varphi &= 2 \cos \varphi \cos n\varphi, \\ \sin(n+1)\varphi + \sin(n-1)\varphi &= 2 \cos \varphi \sin n\varphi, \end{aligned}$$

one obtains for A_n and B_n the recurrences

$$\begin{aligned} A_{n+1}(x) &= xA_n(x) - A_{n-1}(x), \\ B_{n+1}(x) &= xB_n(x) - B_{n-1}(x). \end{aligned} \quad (6)$$

From these, A_n, B_n can be computed for any n , once the functions for $n = 1$ and $n = 2$ are known. Thus one finds

$$\begin{aligned} A_4(x) &= x^4 - 4x^2 + 2, & B_4(x) &= x^3 - 2x, \\ A_5(x) &= x^5 - 5x^3 + 5x, & B_5(x) &= x^4 - 3x^2 + 1, \\ A_6(x) &= x^6 - 6x^4 + 9x^2 - 2, & B_6(x) &= x^5 - 4x^3 + 3x. \end{aligned}$$

and so on. By induction one arrives at the following general law:

$$A_n(x) = \sum_v (-1)^v \frac{n}{n-v} B_v^{(n-v)} x^{n-2v}, \quad 0 \leq v \leq \frac{n}{2}, \quad (7)$$

$$B_n(x) = \sum_v (-1)^v B_v^{(n-v-1)} x^{n-2v-1}, \quad 0 \leq v \leq \frac{n-1}{2}, \quad (8)$$

where the sums begin with $v = 0$ and are continued as long as the exponents of x do not become negative.

Since these formulas prove correct in the first cases, it remains only to show in general that they satisfy (6). For this purpose form

$$A_{n-1}(x) = \sum_v (-1)^v \frac{n-1}{n-v-1} B_v^{(n-v-1)} x^{n-2v-1},$$

and replace v by $v-1$. This gives

$$A_{n-1}(x) = - \sum_v (-1)^v \frac{n-1}{n-v} B_{v-1}^{(n-v)} x^{n-2v+1}, \quad 1 \leq v \leq \frac{n+1}{2}. \quad (9)$$

If we put $B_{-1}^{(n)} = 0$, then v may again be allowed to begin with 0. The values taken by v in (7) differ from those in (9) only when n is odd, because then $v = (n+1):2$ occurs in (9) but not in (7). But this value can be added to (7) if we assume

$$B_{(n+1)/2}^{((n-1)/2)} = 0.$$

It then follows that

$$xA_n(x) - A_{n-1}(x) = \sum_v (-1)^v x^{n-2v+1} \left\{ \frac{n}{n-v} B_v^{(n-v)} + \frac{n-1}{n-v} B_{v-1}^{(n-v)} \right\}. \quad (10)$$

But, as is easily obtained from the expressions for the binomial coefficients (§8),

$$\frac{n}{n-v} B_v^{(n-v)} + \frac{n-1}{n-v} B_{v-1}^{(n-v)} = \frac{n+1}{n-v+1} B_v^{(n-v+1)},$$

including the two boundary cases $v = 0$ and $v = (n+1):2$. Thus (10) agrees with what results from the right side of (7) when n is replaced by $n+1$. In exactly the same way, formula (8) gives

$$xB_n(x) - B_{n-1}(x) = \sum_v (-1)^v x^{n-2v} \left(B_v^{(n-v-1)} + B_{v-1}^{(n-v-1)} \right),$$

and since

$$B_v^{(n-v-1)} + B_{v-1}^{(n-v-1)} = B_v^{(n-v)},$$

(8) is also proved in general.

For clarity we write the formulas somewhat more fully:

$$A_n(x) = x^n - nx^{n-2} + \frac{n(n-3)}{1 \cdot 2} x^{n-4} - \frac{n(n-4)(n-5)}{1 \cdot 2 \cdot 3} x^{n-6} + \dots, \quad (11)$$

$$B_n(x) = x^{n-1} - (n-2)x^{n-3} + \frac{(n-3)(n-4)}{1 \cdot 2} x^{n-5} - \frac{(n-4)(n-5)(n-6)}{1 \cdot 2 \cdot 3} x^{n-7} + \dots. \quad (12)$$

If $\cos n\varphi$ and $\sin n\varphi$ are given, then the equation of degree n

$$2 \cos n\varphi = A_n(x) \quad (13)$$

serves to determine the unknown $x = 2 \cos \varphi$. The meaning of the n roots of this equation follows from the fact that the cosine is unchanged when the angle is increased by a multiple of 2π . Therefore every value

$$x = 2 \cos \left(\varphi + \frac{2v\pi}{n} \right) \quad (14)$$

satisfies (13), where v denotes an arbitrary positive or negative integer. All distinct values of (14) are obtained when v runs through a complete residue system modulo n , for example $v = 0, 1, 2, \dots, n-1$; and one easily sees that these n values are all distinct except in the special case in which φ is a multiple of π . For two cosines are equal only when the sum or the difference of the angles is a multiple of 2π .

The equation

$$\sin n\varphi = \sin \varphi B_n(x) \quad (15)$$

determines $\sin \varphi$ uniquely from x if $B_n(x)$ is not zero.

If in (13) we look for the term independent of x , then using (6) gives, for even n ,

$$2(-1)^{n/2} - 2 \cos n\varphi,$$

and for odd n ,

$$-2 \cos n\varphi.$$

Equating this to the positive or negative product of the roots, one obtains for even n

$$2^{n-1} \prod_{v=0}^{n-1} \cos \left(\varphi + \frac{2v\pi}{n} \right) = (-1)^{n/2} - \cos n\varphi, \quad (16)$$

and for odd n

$$2^{n-1} \prod_{v=0}^{n-1} \cos \left(\varphi + \frac{2v\pi}{n} \right) = \cos n\varphi. \quad (17)$$

Since both sides are continuous functions of φ , these formulas also remain valid in the initially excluded case where φ is a multiple of π .

Under the product sign we may replace v by vm if m is any integer relatively prime to n , since then vm runs through a complete residue system modulo n along with v . Separating off the value $v = 0$, and then letting v run through both the positive and negative integers of absolute value less than $n : 2$, we get from (17), for every odd n ,

$$2^{n-1} \prod_{v=1}^{(n-1)/2} \cos \left(\varphi + \frac{2vm\pi}{n} \right) \cos \left(\varphi - \frac{2vm\pi}{n} \right) = \frac{\cos n\varphi}{\cos \varphi}. \quad (18)$$

Putting $\varphi = 0$ and taking the square root, this gives

$$2^{(n-1)/2} \prod_{v=1}^{(n-1)/2} \cos \frac{2vm\pi}{n} = \pm 1, \quad (19)$$

where the sign remains undetermined for the moment. If, however, in (18) we let φ pass to $\pi : 2$, the right side, which assumes the indeterminate form $0 : 0$, obtains the limiting value $(-1)^{(n-1)/2}n$. Thus we again obtain the formula derived in the preceding paragraph for prime n , now generally for every odd n :

$$2^{(n-1)/2} \prod_{v=1}^{(n-1)/2} \sin \frac{2vm\pi}{n} = \pm \sqrt{n}, \quad (20)$$

and for $m = 1$ the positive sign is to be taken.

The function B_n , which has degree $n - 1$ in x , vanishes when $n\varphi$, but not φ , is a multiple of π ; that is, for

$$x = 2 \cos \frac{h\pi}{n}, \quad (21)$$

where h is an integer not divisible by n . First suppose that n is even. Then we obtain all the distinct values contained in (21) if we put m for any number relatively prime to n and take

$$h = m, 2m, 3m, \dots, (n-1)m. \quad (22)$$

For

$$\cos \frac{h\pi}{n} = \cos \frac{h'\pi}{n}$$

can occur only if either $h + h'$ or $h - h'$ is a multiple of $2n$, which cannot happen when h and h' are both taken from the sequence (22).

Among the values (21), one, namely the one corresponding to $h = mn : 2$, is zero, while the remaining values are pairwise opposite. Hence for even n

$$B_n(x) = x \prod_{v=1}^{(n-2)/2} \left(x^2 - 4 \cos^2 \frac{vm\pi}{n} \right). \quad (23)$$

If n is odd, then $B_n(x)$ depends only on x^2 ; hence two values in (21) are opposite to each other for h and $h + n$. Taking again m relatively prime to n , and putting

$$h = 2m, 4m, 6m, \dots, (n-1)m,$$

one obtains in (21) neither two equal nor two opposite values, and thus all distinct values of x^2 . Therefore, for odd n ,

$$B_n(x) = \prod_{v=1}^{(n-1)/2} \left(x^2 - 4 \cos^2 \frac{2vm\pi}{n} \right). \quad (24)$$

Now put in this formula

$$x = 2 \cos \varphi, \quad x^2 = 4(1 - \sin^2 \varphi), \quad \cos^2 \frac{2vm\pi}{n} = 1 - \sin^2 \frac{2vm\pi}{n}.$$

It can then be written

$$\frac{\sin n\varphi}{\sin \varphi} = (-1)^{(n-1)/2} 2^{n-1} \prod_{v=1}^{(n-1)/2} \left(\sin^2 \varphi - \sin^2 \frac{2vm\pi}{n} \right), \quad (25)$$

from which formula (20) again follows for $\varphi = 0$.

Since $\cos n\varphi$ vanishes for odd n when

$$\cos \varphi = \pm \sin \frac{2vm\pi}{n},$$

we have thereby also found the factorization of the function $A_n(x)$ in this case. For odd n ,

$$A_n(x) = x \prod_{v=1}^{(n-1)/2} \left(x^2 - 4 \sin^2 \frac{2vm\pi}{n} \right). \quad (26)$$

The quantities

$$\omega = \sin^2 \frac{2vm\pi}{n} \quad (27)$$

are therefore the roots of an algebraic equation of degree $(n-1) : 2$,

$$\Phi_n(x) = 0.$$

We obtain the function $\Phi_n(x)$ by replacing the variable x^2 in the function $A_n(x) : x$ by a new variable $4x$ and dividing by 2^{n-1} . Hence [formula (11)]

$$\begin{aligned} \Phi_n(x) &= \prod (x - \omega) \\ &= x^{(n-1)/2} - \frac{n}{4} x^{(n-3)/2} + \frac{n(n-3)}{1 \cdot 2 \cdot 16} x^{(n-5)/2} - \dots, \end{aligned} \quad (28)$$

and (25) can be written

$$\frac{\sin n\varphi}{\sin \varphi} = (-1)^{(n-1)/2} 2^{n-1} \prod (\sin^2 \varphi - \omega). \quad (29)$$